$$\tilde{\psi}(p) = \langle \mathbf{P} | \Psi \rangle. \qquad (8.20)$$

It is now clear that there are two ways to represent a state-vector. One way is in the position basis and the other is in the momentum basis. Both wave functions—the position wave function $\psi(x)$ and the momentum wave function $\tilde{\psi}(p)$—represent exactly the same state-vector $|\Psi\rangle$. It follows that there must be some transformation between them such that if you know $\psi(x)$, the transformation produces $\tilde{\psi}(p)$, and vice versa. In fact, the two representations are Fourier transforms of each other.

8.3.1 Resolving the Identity

We are about to see the great power of the Dirac bra-ket notation in simplifying complicated things. First, let's recall an important idea from earlier lectures. Suppose we define an orthonormal basis of states through the eigenvectors of some Hermitian observable. Call the basis vectors $|i\rangle$. In Lecture 7, I explained a very useful trick, and now we are going to see just how useful it is. It's called *resolving the identity*. The trick given in (Eq. 7.11) is to write the identity operator I (the operator that acts on any vector to give the same vector) in the form

$$\mathbf{I} = \sum_i |i\rangle\langle i|.$$

Because momentum and position are both Hermitian, the sets of vectors $|x\rangle$ and set $|p\rangle$ each define basis vectors. By replacing summation with integration we discover two ways to resolve the identity:

$$\mathbf{I} = \int dx |x\rangle\langle x| \qquad (8.21)$$

and